

## FRIDAY HARBOR, WA, USA

WMO: 727985

Lat: 48.522N

Lon: 123.023W

Elev: 109

StdP: 14.64

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 94276

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | 26.4       | 29.3 | 9.6                           | 9.0 | 30.9 | 15.9 | 12.3 | 33.1 | 30.7                  | 46.0 | 27.7 | 45.7 | 4.8                   | 10   | 0.547 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.1                   | 79.5            | 62.8 | 75.3 | 61.6 | 72.2 | 60.4 | 64.0                | 77.2 | 62.3 | 73.8 | 61.0 | 71.0 | 6.4                  | 10   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 58.5                            | 73.5 | 66.6 | 57.2 | 70.0 | 65.5 | 56.1 | 67.2 | 64.4 | 29.1          | 77.2 | 28.0 | 73.9 | 27.0 | 70.8 | 71.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 23.6              | 18.7 | 15.5 | 19.8                       | 87.7 | 4.8                | 4.1 | 16.4                                               | 90.7 | 13.6       | 93.1 | 10.9       | 95.4 | 7.5        | 98.4 |
|                   |      |      | 17.9                       | 67.4 | 4.4                | 1.5 | 14.7                                               | 68.5 | 12.2       | 69.4 | 9.7        | 70.2 | 6.5        | 71.3 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 50.1  | 41.6  | 41.6  | 44.6  | 48.2  | 53.0  | 57.3  | 60.4  | 60.8  | 57.3  | 50.6  | 44.8  | 40.9 | (6)  |
| (7)  |                                                                                  | DBStd       | 8.43  | 5.20  | 4.91  | 4.45  | 4.05  | 4.12  | 3.91  | 3.79  | 3.45  | 3.69  | 3.95  | 5.20  | 5.23 | (7)  |
| (8)  |                                                                                  | HDD50       | 1255  | 262   | 236   | 173   | 79    | 16    | 1     | 0     | 0     | 0     | 40    | 166   | 282  | (8)  |
| (9)  |                                                                                  | HDD65       | 5446  | 726   | 655   | 632   | 503   | 371   | 236   | 153   | 139   | 233   | 447   | 605   | 746  | (9)  |
| (10) |                                                                                  | CDD50       | 1308  | 1     | 1     | 6     | 26    | 110   | 219   | 322   | 334   | 219   | 58    | 11    | 1    | (10) |
| (11) |                                                                                  | CDD65       | 26    | 0     | 0     | 0     | 0     | 0     | 5     | 10    | 9     | 2     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH74       | 497   | 0     | 0     | 0     | 1     | 11    | 86    | 192   | 169   | 38    | 0     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH80       | 99    | 0     | 0     | 0     | 0     | 1     | 18    | 46    | 30    | 4     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 5.3   | 6.4   | 6.2   | 6.3   | 5.5   | 4.8   | 4.9   | 4.5   | 3.9   | 3.7   | 4.6   | 6.3   | 6.9  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 33.3  | 5.0   | 3.1   | 3.0   | 1.9   | 1.4   | 1.2   | 0.7   | 1.1   | 1.6   | 3.7   | 5.3   | 5.2  | (15) |
| (16) |                                                                                  | PrecMax     | 42.0  | 8.4   | 7.4   | 5.7   | 3.8   | 2.9   | 2.5   | 1.6   | 3.6   | 4.7   | 10.7  | 11.1  | 8.2  | (16) |
| (17) |                                                                                  | PrecMin     | 24.7  | 2.3   | 0.5   | 0.9   | 0.2   | 0.1   | 0.2   | 0.0   | 0.2   | 0.1   | 0.6   | 1.9   | 2.0  | (17) |
| (18) |                                                                                  | PrecStd     | 4.7   | 1.6   | 1.4   | 1.1   | 0.9   | 0.7   | 0.6   | 0.4   | 0.9   | 1.2   | 2.1   | 2.4   | 1.7  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 54.3  | 54.3  | 61.3  | 68.1  | 74.8  | 82.7  | 85.2  | 83.7  | 79.0  | 65.9  | 59.1  | 54.2 | (19) |
| (20) |                                                                                  |             | MCWB  | 49.7  | 47.2  | 51.2  | 54.7  | 59.4  | 62.6  | 65.2  | 64.2  | 61.5  | 54.9  | 53.1  | 50.2 | (20) |
| (21) |                                                                                  | 2%          | DB    | 52.0  | 52.1  | 56.3  | 62.8  | 69.7  | 75.4  | 79.1  | 78.8  | 73.1  | 62.4  | 55.6  | 51.6 | (21) |
| (22) |                                                                                  |             | MCWB  | 48.6  | 46.6  | 48.7  | 52.3  | 57.5  | 60.9  | 63.3  | 63.0  | 59.6  | 53.9  | 52.0  | 48.4 | (22) |
| (23) |                                                                                  | 5%          | DB    | 50.2  | 50.1  | 53.9  | 58.8  | 65.3  | 70.4  | 74.6  | 74.5  | 69.6  | 59.5  | 53.9  | 49.5 | (23) |
| (24) |                                                                                  |             | MCWB  | 47.1  | 45.7  | 47.5  | 50.1  | 55.7  | 59.5  | 61.7  | 61.9  | 58.6  | 53.3  | 50.8  | 46.3 | (24) |
| (25) |                                                                                  | 10%         | DB    | 48.4  | 48.3  | 51.8  | 55.9  | 62.3  | 66.3  | 71.1  | 71.3  | 66.0  | 57.2  | 52.1  | 47.8 | (25) |
| (26) |                                                                                  |             | MCWB  | 45.5  | 44.7  | 46.6  | 48.7  | 54.4  | 57.3  | 60.5  | 60.7  | 57.5  | 52.3  | 49.2  | 45.0 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 51.9  | 50.5  | 52.8  | 56.3  | 60.9  | 64.2  | 66.8  | 65.5  | 63.4  | 58.6  | 54.9  | 51.9 | (27) |
| (28) |                                                                                  |             | MCDWB | 52.8  | 52.4  | 58.1  | 65.7  | 71.7  | 78.3  | 82.5  | 79.7  | 74.1  | 62.1  | 56.7  | 53.1 | (28) |
| (29) |                                                                                  | 2%          | WB    | 49.5  | 48.1  | 50.1  | 53.2  | 58.5  | 61.7  | 64.1  | 63.8  | 61.2  | 56.0  | 53.1  | 48.7 | (29) |
| (30) |                                                                                  |             | MCDWB | 51.2  | 50.4  | 54.3  | 60.8  | 67.8  | 74.3  | 77.8  | 76.6  | 70.4  | 59.1  | 55.1  | 50.4 | (30) |
| (31) |                                                                                  | 5%          | WB    | 47.7  | 46.5  | 48.5  | 51.2  | 56.5  | 59.8  | 62.2  | 62.3  | 59.6  | 54.4  | 51.2  | 46.8 | (31) |
| (32) |                                                                                  |             | MCDWB | 49.4  | 48.8  | 52.4  | 57.2  | 64.2  | 69.4  | 73.7  | 73.4  | 67.0  | 57.6  | 53.2  | 48.5 | (32) |
| (33) |                                                                                  | 10%         | WB    | 46.2  | 45.6  | 47.2  | 49.8  | 54.8  | 58.1  | 60.6  | 61.0  | 58.2  | 53.2  | 49.6  | 45.7 | (33) |
| (34) |                                                                                  |             | MCDWB | 47.9  | 47.8  | 50.9  | 54.7  | 61.4  | 65.9  | 70.1  | 70.4  | 64.8  | 56.3  | 51.6  | 47.3 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 8.6   | 10.6  | 12.5  | 15.1  | 17.4  | 18.3  | 21.1  | 21.1  | 17.9  | 12.8  | 9.8   | 8.4  | (35) |
| (36) |                                                                                  |             | MCDBR | 9.2   | 11.9  | 14.8  | 19.8  | 23.9  | 26.2  | 28.6  | 27.6  | 24.0  | 15.7  | 9.7   | 8.5  | (36) |
| (37) |                                                                                  |             | MCWBR | 6.8   | 8.0   | 9.2   | 11.2  | 13.5  | 13.6  | 14.1  | 13.6  | 12.1  | 9.5   | 7.0   | 6.6  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 7.9   | 9.5   | 13.2  | 17.2  | 20.8  | 23.3  | 25.9  | 24.7  | 20.2  | 12.8  | 8.6   | 7.8  | (38) |
| (39) |                                                                                  |             | MCWBR | 6.7   | 7.1   | 8.9   | 10.5  | 12.4  | 12.9  | 13.5  | 12.9  | 11.2  | 8.7   | 6.8   | 6.6  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.318 | 0.316 | 0.330 | 0.355 | 0.377 | 0.355 | 0.348 | 0.354 | 0.340 | 0.331 | 0.323 | 0.313 | (40) |      |
| (41) |                                                                                  | taud        | 2.529 | 2.558 | 2.533 | 2.453 | 2.407 | 2.487 | 2.538 | 2.521 | 2.559 | 2.569 | 2.524 | 2.499 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 240   | 270   | 283   | 284   | 281   | 287   | 288   | 282   | 276   | 260   | 235   | 226   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 19    | 23    | 27    | 33    | 36    | 33    | 31    | 30    | 26    | 23    | 19    | 17    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 293   | 574   | 897   | 1361  | 1742  | 1905  | 2080  | 1753  | 1247  | 685   | 351   | 235   | (44) |      |
| (45) |                                                                                  | RadStd      | 39    | 77    | 105   | 127   | 137   | 180   | 146   | 114   | 102   | 76    | 54    | 31    | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |      |
| (46) | Station Only            | +0.81 | N/A     | -5.47  | +1.76   | +0.74 | N/A   | N/A         | -340  | +274  | +14   | (46) |
| (47) | Regional (74 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   | (47) |

Nomenclature: See separate page